

Yuanyi Pei

yunayi.pei@icahn.mssm.edu | +86 18601186220 | Huangqu East Road, Beijing, China

Education

Icahn School of Medicine at Mount Sinai

New York, NY, USA

MS Candidate in Biomedical Sciences | GPA: 3.91/4.0

Aug. 2025 – Jun.2027(Expected)

Core courses: Neuroscience, Neuropharmacology, Physiology & Biophysics, Biostatistics, Neural Basis of Neurological & Psychiatric Disorders (In Progress), Seminar in Neuroscience (In Progress)

China Agricultural University

Beijing, China

BSc in Biological Sciences | GPA: 3.62/4.0

Sept. 2021 – Jun. 2025

Core courses: Neurobiology, Molecular Biology, Cell Biology, Genetics, Animal Physiology, Biochemistry, Microbiology, Genetics Laboratory, Developmental Biology

Graduation Thesis: *Identification of the sleep regulatory center using retrograde tracing from the orbicularis oculi muscle*

Research Interests

Sleep neurobiology, **hypothalamic and orexinergic circuits**, NREM-dependent memory processing, and sleep dysfunction in Alzheimer's disease.

Publications

Pei, Y., Witztum, R., Lozano, K., Kam, K., & Varga, A. (2026). 0054 Effects of Sik3 Sleepy Mutation on Hypothalamic Function in Mice. *Sleep*, 49 (Supplement_1), A24-A25. <https://doi.org/10.1093/sleep/zsag091.0054>

Pei, Y., Ohmura, Y., & Yamanaka, A. Compact human prepro-orexin promoters for selective AAV₂/PHP.eB-mediated targeting of orexin/hypocretin neurons (Submitted to *Cell Reports Methods*)

Conference Presentations

Witztum, R., Lozano, K., **Pei, Y. (Co-first author)**, Kam, K., & Varga, A. (2026). *Effects of Sik3 sleepy mutation on hypothalamic function in mice*. SLEEP 2026, San Diego, CA. (Oral and poster presentation)

Projects Involved

Effects of Sik3 Sleepy Mutation on Hypothalamic Function in Mice

Jun 2025 – Present

Advisor: Dr. Andrew Varga

Project Introduction: Although traditionally considered an ideal idiopathic hypersomnia model, the Sik3 sleepy mutant shows problems resembling an NT2 model. We give mice chocolate to test cataplexy induction, observe a symptom termed DT sleep, and find that Sik3 mice show hypothalamic dysfunction (e.g., abnormal feeding). Therefore, we are injecting a virus containing the SIK3 mutant into the hypothalamus of mice.

- Mastering EEG/EMG signal analysis to analyze recorded EEG/EMG data
- Using chocolate, an easily accessible and widely used method for detecting cataplexy, to assess behavioral abnormalities in Sik3 mice
- Using curcumin to attempt to inhibit SIK3 activity in mice and performing oral gavage for administration
- Assisting the lab in completing a large number of paraffin sections and establishing a cryosectioning protocol for the lab

Optimization of the Human Prepro-Orexin Promoter for Selective Targeting of Orexin/Hypocretin Neurons Using AAV-PHP.eB

Jul 2024 – Aug 2025

❖ Chinese Institute for Brain Research (CIBR), Beijing, China

Advisor: Dr. Akihiro Yamanaka

- Designed truncated variants of the human prepro-orexin promoter through comprehensive literature review and rational design, aiming not to enhance promoter efficiency but to reduce the originally oversized promoter from 3.2 kb to 2.2 kb
- Validated promoter performance through both systemic delivery (retro-orbital injection) for whole-brain neuronal targeting and stereotaxic injection into the lateral hypothalamus for high-resolution axonal projection mapping
- Identified a minimal promoter construct (>96% coverage, >97% specificity) that achieves stable, neuron-specific transgene expression while overcoming the size limitation of the full-length promoter; further validated that the construct drives jGCaMP8S protein expression and is compatible with miniaturized two-photon (Mini2P)-related experiments

Undergraduate Thesis: Identification of the Sleep Regulatory Center Using Retrograde Tracing from the Orbicularis Oculi Muscle

Nov 2024 – Jun 2025

Advisor: Dr. Akihiro Yamanaka

- Motivated by clinical observations that human eyelids remain closed during sleep, yet some patients exhibit eyelid non-closure during excessive alcohol intake, and people feel eyelid weakness under high sleep pressure, suggesting a potential sleep-related pathway.
- Designed surgical protocols and selected retrograde tracers such as CTB for precise injections into the orbicularis oculi muscle in mice
- Established a retrograde tracing pipeline to map central neural circuits underlying sleep-related eyelid closure
- Identified a subpopulation of cholinergic (ChAT⁺) motor neurons in the lateral/peripheral region of the facial nucleus (7N) and found preliminary evidence suggesting hypothalamic inputs to these neurons
- Aimed to elucidate neural mechanisms of nocturnal lagophthalmos and its potential link to hypothalamic sleep centers

Research Training

Andrew Varga Lab, Icahn School of Medicine at Mount Sinai, New York, NY, USA

Jun 2025 – Present

Graduate Research Assistant

- Currently preparing for thesis research on sleep neurobiology; finalizing experimental design and literature review in anticipation of in vivo electrophysiology and behavioral experiments

Akihiro Yamanaka & Yu Ohmura Labs, Chinese Institute for Brain Research (CIBR), Beijing, China

Oct 2022 – Jul 2025

Research Intern

- Conducted stereotaxic delivery of AAV vectors into hypothalamic regions and implanted optical fibers for in vivo manipulation and recording of orexinergic neurons
- Engineered truncated variants of the human prepro-orexin promoter to improve targeting specificity and overcome AAV packaging constraints
- Performed cardiac perfusion, cryosectioning, immunohistochemistry, and fluorescence microscopy to validate transgene expression and neural projections
- Assisted in behavioral assay setup and preprocessing for sleep-related experiments; familiar with EEG recording principles and analysis workflows
- Participated in lab meetings and presented at internal seminars; attended Cold Spring Harbor Asia conference on sleep neurobiology

Chen Liangyi Lab, Peking University, Beijing, China

Jun 2024 – Oct 2024

Undergraduate Research Intern

- Performed stereotaxic and cranial window surgeries for in vivo two-photon and wide-field single-photon imaging in awake mice
- Processed calcium imaging data using CalmAn pipeline, implementing CNMF and motion correction algorithms to extract neural activity traces
- Analyzed large field-of-view neural dynamics during behavioral tasks using custom Python scripts, enhancing coding proficiency in scientific computing
- Participated in lab meetings and training sessions on optical imaging principles, experimental design, and data interpretation

Technical Skills

- **In Vivo Techniques:** Stereotaxic surgery, fiber photometry implantation, systemic injections (i.p./retro-orbital/tail vein), and cranial window surgery for in vivo two-photon imaging; familiar with EEG recording principles
- **Viral & Molecular Tools:** AAV handling, plasmid construction (restriction-ligation, Gateway cloning), SnapGene, mini/maxi-prep, colony PCR, high-fidelity PCR (PrimeSTAR)
- **Histology & Imaging:** Cardiac perfusion, cryosectioning, immunohistochemistry, fluorescence/confocal microscopy, ClearMap
- **Data Analysis:** Python (CaImAn, CNMF, NumPy, SciPy), SPSS, Origin; neural activity extraction, motion correction, ROI analysis
- **Behavioral Methods:** Pre-handling and data analysis for novel object recognition and other basic behavioral assays
- **Scientific Communication:** Scientific figure design (BioRender), poster presentation, literature synthesis